# Partial Progress on Robustness of Approximate Controlled Diffusion Models in Degenerate Limits

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**Abstract**

Pradhan and Yüksel prove value convergence and vanishing model-mismatch loss for approximate controlled diffusion models under several cost criteria in a nondegenerate setting. Their work uses regularity of uniformly elliptic Hamilton-Jacobi-Bellman equations, and a natural open direction is to understand what remains true when the limiting true diffusion is degenerate. This note gives partial progress, not a solution of the full problem. Under common global Lipschitz and growth estimates, locally uniform convergence of coefficients and bounded costs, a direct coupling proof gives compact-uniform finite-horizon and discounted robustness for open-loop controls. The same argument also gives robustness for uniformly regular feedback classes. No ellipticity, hypoellipticity, or smoothing of the limiting diffusion is used. We also formulate an average-cost lifting theorem under an explicit uniform finite-time cost convergence and uniform ergodic-stability assumption. Finally, a simple deterministic counterexample shows why finite-time state convergence alone cannot handle arbitrary discontinuous feedback selectors; this is the obstruction preventing the present argument from solving the full Pradhan-Yüksel problem.

## 1 Status relative to the Pradhan–Yüksel problem

The results below should be read as partial progress. They prove that ellipticity is not needed for certain robustness conclusions once one restricts to model-independent open-loop controls, or to feedback classes with enough uniform regularity. They do not solve the full robustness problem from [1, 2]. The missing pieces are substantive:

(i) the argument does not treat arbitrary measurable optimal Markov or stationary Markov selectors;

(ii) it does not prove existence or stability of optimal selectors for degenerate Hamilton-Jacobi-Bellman equations;

(iii) it does not treat the exit-time criterion;

(iv) the average-cost result below assumes a uniform long-run stability hypothesis rather than deriving it from degenerate-control structure;

(v) the coefficient convergence assumptions are stronger than those in the nondegenerate theory of [1, 2].

The positive message is nevertheless useful: the finite-time and discounted cost-continuity mechanism itself is probabilistic and does not require a lower ellipticity bound. The hard missing issue is policy regularity/selection, not the convergence of coupled state processes under the same sufficiently regular policy.

## 2 Controlled diffusions and assumptions

Let $A$ be a compact metric action space, and let $d, m \in \mathbb{N}$. For $n \geq 0$, let

$$b_n : \mathbb{R}^d \times A \to \mathbb{R}^d, \qquad \sigma_n : \mathbb{R}^d \times A \to \mathbb{R}^{d\times m}, \qquad c_n : \mathbb{R}^d \times A \to \mathbb{R}$$

be, respectively, the drift, diffusion coefficient, and running cost of model $n$. The model $n = 0$ is the limiting true model. No lower bound is imposed on $\sigma_0\sigma_0^\top$; in particular $\sigma_0 \equiv 0$ is allowed.

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfy the usual conditions and carry an $m$-dimensional Brownian motion $W$. Let $\mathcal{U}$ denote the set of all $A$-valued progressively measurable controls. For $u \in \mathcal{U}$ and $x \in \mathbb{R}^d$, let $X^{n,x,u}$ solve

$$dX_t^{n,x,u} = b_n(X_t^{n,x,u}, u_t)\, dt + \sigma_n(X_t^{n,x,u}, u_t)\, dW_t, \qquad X_0^{n,x,u} = x. \tag{1}$$

We use the following assumptions throughout Sections 3 and 4.

**Assumption 2.1** (Basic open-loop hypotheses)**.** *The following hold.*

*(A1) The functions $b_n, \sigma_n, c_n$ are Borel measurable, and $c_0$ is continuous on $\mathbb{R}^d \times A$.*

*(A2) There is $L < \infty$ such that, for all $n \geq 0$, $x, y \in \mathbb{R}^d$, and $a \in A$,*

$$|b_n(x,a) - b_n(y,a)| + \|\sigma_n(x,a) - \sigma_n(y,a)\| \leq L\,|x-y|\,,$$

$$|b_n(x,a)| + \|\sigma_n(x,a)\| \leq L(1 + |x|).$$

*Here $\|\cdot\|$ is the Hilbert–Schmidt norm.*

*(A3) For every $R < \infty$,*

$$\delta_n(R) := \sup_{|x|\leq R,\, a\in A} \big(|b_n(x,a) - b_0(x,a)| + \|\sigma_n(x,a) - \sigma_0(x,a)\| + |c_n(x,a) - c_0(x,a)|\big) \longrightarrow 0.$$

*(A4) The running costs are uniformly bounded: there is $M < \infty$ such that*

$$\sup_{n\geq 0} \sup_{x\in\mathbb{R}^d,\, a\in A} |c_n(x,a)| \leq M.$$

Under Assumption 2.1, (1) has a unique strong solution for every $n, x, u$, since the random coefficients $(t, \omega, x) \mapsto b_n(x, u_t(\omega))$ and $(t, \omega, x) \mapsto \sigma_n(x, u_t(\omega))$ are progressively measurable, globally Lipschitz in $x$, and satisfy a linear growth bound uniformly in $(t, \omega)$.

For the finite-horizon criterion we also use terminal costs $g_n : \mathbb{R}^d \to \mathbb{R}$.

**Assumption 2.2** (Terminal costs)**.** *There is $M_g < \infty$ such that $\sup_{n,x} |g_n(x)| \leq M_g$. The function $g_0$ is continuous, and for every $R < \infty$,*

$$\sup_{|x|\leq R} |g_n(x) - g_0(x)| \longrightarrow 0.$$

# 3 Finite-horizon robustness

For $T < \infty$, define

$$J_{T,n}(x,u) := \mathbb{E}\left[\int_0^T c_n(X_t^{n,x,u}, u_t)\, dt + g_n(X_T^{n,x,u})\right], \qquad V_{T,n}(x) := \inf_{u\in\mathcal{U}} J_{T,n}(x,u).$$

**Lemma 3.1** (Uniform moment bound)**.** *Assume (A2). For every compact $K \subset \mathbb{R}^d$ and $T < \infty$ there is $C_{K,T} < \infty$ such that*

$$\sup_{n\geq 0}\sup_{x\in K}\sup_{u\in\mathcal{U}} \mathbb{E}\left[\sup_{0\leq t\leq T} |X_t^{n,x,u}|^2\right] \leq C_{K,T}.$$

*Proof.* For $t \leq T$, the elementary inequality $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$, the linear growth condition, Cauchy's inequality, and the Burkholder-Davis-Gundy inequality give

$$\begin{aligned}\mathbb{E}\sup_{s\leq t}|X_s^{n,x,u}|^2 &\leq C\,|x|^2 + C\mathbb{E}\sup_{s\leq t}\left|\int_0^s b_n(X_r^{n,x,u}, u_r)\, dr\right|^2 + C\mathbb{E}\sup_{s\leq t}\left|\int_0^s \sigma_n(X_r^{n,x,u}, u_r)\, dW_r\right|^2 \\ &\leq C_T(1+|x|^2) + C_T\int_0^t \mathbb{E}\sup_{q\leq r}\left|X_q^{n,x,u}\right|^2\, dr,\end{aligned}$$

where $C_T$ depends only on $L, T, d, m$. Gronwall's lemma and compactness of $K$ yield the assertion. □

**Lemma 3.2** (Uniform finite-time state convergence)**.** *Assume (A2) and (A3). For every compact $K \subset \mathbb{R}^d$, every $T < \infty$, and every $\eta > 0$,*

$$\sup_{x\in K}\sup_{u\in\mathcal{U}} \mathbb{P}\left(\sup_{0\leq t\leq T}\left|X_t^{n,x,u} - X_t^{0,x,u}\right| > \eta\right) \longrightarrow 0.$$

*Proof.* Fix $K, T$, and let $R > 0$. Set

$$\tau_R^{n,x,u} := \inf\{t \geq 0 :\ |X_t^{n,x,u}| \vee \left|X_t^{0,x,u}\right| \geq R\}$$

and $\Delta_t = X_t^{n,x,u} - X_t^{0,x,u}$. On $[0, T\wedge\tau_R^{n,x,u}]$, the coefficient mismatch is bounded by $\delta_n(R)$. Hence, by the Lipschitz estimate and the Burkholder-Davis-Gundy inequality, for $t \leq T$,

$$\begin{aligned}\mathbb{E}\sup_{s\leq t\wedge\tau_R^{n,x,u}}|\Delta_s|^2 &\leq C_T\mathbb{E}\int_0^{t\wedge\tau_R^{n,x,u}}\left(|\Delta_r|^2 + \delta_n(R)^2\right) dr \\ &\leq C_T\int_0^t \mathbb{E}\sup_{q\leq r\wedge\tau_R^{n,x,u}}|\Delta_q|^2\, dr + C_T\delta_n(R)^2.\end{aligned}$$

Gronwall's lemma gives

$$\sup_{x\in K}\sup_{u\in\mathcal{U}}\mathbb{E}\sup_{0\leq t\leq T\wedge\tau_R^{n,x,u}}|\Delta_t|^2 \leq C_T\delta_n(R)^2. \tag{2}$$

By Lemma 3.1,

$$\sup_{n\geq 0}\sup_{x\in K}\sup_{u\in\mathcal{U}}\mathbb{P}(\tau_R^{n,x,u} \leq T) \leq \frac{2C_{K,T}}{R^2}. \tag{3}$$

Therefore

$$\sup_{x\in K}\sup_{u\in\mathcal{U}}\mathbb{P}\left(\sup_{t\leq T}|\Delta_t| > \eta\right) \leq \frac{C_T\delta_n(R)^2}{\eta^2} + \frac{2C_{K,T}}{R^2}.$$

Let $n \to \infty$ with $R$ fixed, and then let $R \to \infty$. □

**Lemma 3.3** (Uniform finite-horizon cost convergence)**.** *Assume 2.1 and 2.2. For every compact $K \subset \mathbb{R}^d$ and every $T < \infty$,*

$$\sup_{x \in K} \sup_{u \in \mathcal{U}} |J_{T,n}(x,u) - J_{T,0}(x,u)| \longrightarrow 0.$$

*Proof.* Fix $K, T$. We prove convergence of the running-cost term; the terminal term is identical with $c_n, c_0, M$ replaced by $g_n, g_0, M_g$ and time $T$ fixed.

Let

$$\Gamma_R^{n,x,u} := \left\{ \sup_{0 \le t \le T} |X_t^{n,x,u}| \vee \left| X_t^{0,x,u} \right| \le R \right\}.$$

By Lemma 3.1,

$$\sup_{n, x \in K, u \in \mathcal{U}} \mathbb{P}\big( (\Gamma_R^{n,x,u})^c \big) \le \frac{2C_{K,T}}{R^2}. \tag{4}$$

The contribution of $(\Gamma_R^{n,x,u})^c$ to the expected absolute running-cost difference is therefore bounded by $4MTC_{K,T}/R^2$, uniformly in $n, x, u$.

On $\Gamma_R^{n,x,u}$,

$$\left| c_n(X_t^{n,x,u}, u_t) - c_0(X_t^{0,x,u}, u_t) \right| \le \delta_n(R) + \omega_R\Big( \left| X_t^{n,x,u} - X_t^{0,x,u} \right| \Big),$$

where

$$\omega_R(r) := \sup_{\substack{|y| \le R,\ |z| \le R,\ |y-z| \le r \\ a \in A}} |c_0(y,a) - c_0(z,a)|.$$

Since $B_R \times A$ is compact and $c_0$ is continuous, $\omega_R(r) \downarrow 0$ as $r \downarrow 0$. For any $\eta > 0$,

$$\omega_R(\left| X_t^{n,x,u} - X_t^{0,x,u} \right|) \le \omega_R(\eta) + 2M \mathbf{1}_{\{\sup_{s \le T} |X_s^{n,x,u} - X_s^{0,x,u}| > \eta\}}.$$

Using Lemma 3.2, we get

$$\limsup_{n \to \infty} \sup_{x \in K, u \in \mathcal{U}} \mathbb{E} \int_0^T \left| c_n(X_t^{n,x,u}, u_t) - c_0(X_t^{0,x,u}, u_t) \right| dt \le \frac{4MTC_{K,T}}{R^2} + T\omega_R(\eta).$$

Let $\eta \downarrow 0$ and then $R \to \infty$. The terminal-cost convergence follows in the same way, using Assumption 2.2 and the modulus of continuity of $g_0$ on $B_R$. □

**Theorem 3.4** (Finite-horizon open-loop robustness)**.** *Assume 2.1 and 2.2. For every compact $K \subset \mathbb{R}^d$ and every $T < \infty$,*

$$\sup_{x \in K} |V_{T,n}(x) - V_{T,0}(x)| \longrightarrow 0.$$

*Moreover, if $x \in \mathbb{R}^d$ is fixed and $u_n \in \mathcal{U}$ satisfies*

$$J_{T,n}(x, u_n) \le V_{T,n}(x) + \varepsilon_n, \qquad \varepsilon_n \downarrow 0,$$

*then the same open-loop control sequence is asymptotically optimal for the degenerate true model:*

$$J_{T,0}(x, u_n) - V_{T,0}(x) \longrightarrow 0.$$

*Proof.* The value convergence follows from Lemma 3.3 and

$$\left| \inf_{u\in\mathcal{U}} f_n(u) - \inf_{u\in\mathcal{U}} f_0(u) \right| \le \sup_{u\in\mathcal{U}} |f_n(u) - f_0(u)| \,.$$

For robustness, set

$$\eta_n(x) := \sup_{u\in\mathcal{U}} |J_{T,n}(x,u) - J_{T,0}(x,u)| \,.$$

Then

$$\begin{aligned} 0 \le J_{T,0}(x,u_n) - V_{T,0}(x) &\le |J_{T,0}(x,u_n) - J_{T,n}(x,u_n)| + \big(J_{T,n}(x,u_n) - V_{T,n}(x)\big) \\ &\quad + |V_{T,n}(x) - V_{T,0}(x)| \\ &\le 2\eta_n(x) + \varepsilon_n \longrightarrow 0. \end{aligned}$$

□

# 4 Discounted robustness

For $\alpha > 0$, define

$$J_{\alpha,n}(x,u) := \mathbb{E}\left[\int_0^\infty e^{-\alpha t} c_n(X_t^{n,x,u}, u_t)\, dt\right], \qquad V_{\alpha,n}(x) := \inf_{u\in\mathcal{U}} J_{\alpha,n}(x,u).$$

**Theorem 4.1** (Discounted open-loop robustness)**.** *Assume 2.1. For every compact $K \subset \mathbb{R}^d$ and every $\alpha > 0$,*

$$\sup_{x\in K}\sup_{u\in\mathcal{U}} |J_{\alpha,n}(x,u) - J_{\alpha,0}(x,u)| \longrightarrow 0, \qquad \sup_{x\in K} |V_{\alpha,n}(x) - V_{\alpha,0}(x)| \longrightarrow 0.$$

*If $x \in \mathbb{R}^d$ is fixed and $u_n \in \mathcal{U}$ satisfies*

$$J_{\alpha,n}(x,u_n) \le V_{\alpha,n}(x) + \varepsilon_n, \qquad \varepsilon_n \downarrow 0,$$

*then*

$$J_{\alpha,0}(x,u_n) - V_{\alpha,0}(x) \longrightarrow 0.$$

*Proof.* Fix $S < \infty$. The running-cost part of Lemma 3.3, applied on $[0,S]$, gives

$$\sup_{x\in K, u\in\mathcal{U}} \mathbb{E}\int_0^S e^{-\alpha t} \left| c_n(X_t^{n,x,u}, u_t) - c_0(X_t^{0,x,u}, u_t) \right| dt \longrightarrow 0.$$

The tail is bounded uniformly by

$$\sup_{n,x,u} \mathbb{E}\int_S^\infty e^{-\alpha t} \left| c_n(X_t^{n,x,u}, u_t) - c_0(X_t^{0,x,u}, u_t) \right| dt \le \frac{2M}{\alpha} e^{-\alpha S}.$$

Let $n \to \infty$ and then $S \to \infty$. The value and robustness conclusions follow exactly as in Theorem 3.4. □

## 5 Regular feedback classes

The preceding theorems compare the two models under the same exogenous progressively measurable action process $u$. Feedback policies require more care, because model $n$ applies $v(t, X_t^n)$ whereas model 0 applies $v(t, X_t^0)$; these are generally different action processes. The same proof works for uniformly regular feedback classes.

In this section assume $A \subset \mathbb{R}^q$ is compact. Suppose that, in addition to the earlier assumptions, $b_n$ and $\sigma_n$ are uniformly Lipschitz in both state and action: for some $L_a < \infty$,

$$\left|b_n(x,a) - b_n(y,a')\right| + \left\|\sigma_n(x,a) - \sigma_n(y,a')\right\| \le L_a(|x-y| + \left|a-a'\right|)$$

for all $n, x, y, a, a'$. Let $\Lambda < \infty$ and let $\mathfrak{V}_\Lambda$ be a class of Borel feedbacks $v : [0,\infty) \times \mathbb{R}^d \to A$ satisfying

$$|v(t,x) - v(t,y)| \le \Lambda\,|x-y| \qquad \text{for all } t \ge 0,$$

with the same $\Lambda$ for every $v \in \mathfrak{V}_\Lambda$. For $v \in \mathfrak{V}_\Lambda$, write $X^{n,x,v}$ for the closed-loop process

$$dX_t^{n,x,v} = b_n(X_t^{n,x,v}, v(t, X_t^{n,x,v}))\,dt + \sigma_n(X_t^{n,x,v}, v(t, X_t^{n,x,v}))\,dW_t.$$

**Corollary 5.1** (Finite-horizon and discounted robustness over regular feedbacks)**.** *Under the assumptions of this section, Theorems 3.4 and 4.1 remain true with $\mathcal{U}$ replaced by $\mathfrak{V}_\Lambda$ and with the natural closed-loop costs. In particular, values restricted to $\mathfrak{V}_\Lambda$ converge compact-uniformly, and $\varepsilon_n$-optimal regular feedbacks for the approximate models are asymptotically optimal for the true model within the same regular-feedback class.*

*Proof.* For $v \in \mathfrak{V}_\Lambda$, define closed-loop coefficients

$$B_{n,v}(t,x) := b_n(x, v(t,x)), \qquad \Sigma_{n,v}(t,x) := \sigma_n(x, v(t,x)).$$

They satisfy a common Lipschitz bound in $x$ with constant $L_a(1+\Lambda)$ and a common linear growth bound. On $|x| \le R$,

$$|B_{n,v}(t,x) - B_{0,v}(t,x)| + \|\Sigma_{n,v}(t,x) - \Sigma_{0,v}(t,x)\| \le \delta_n(R),$$

uniformly in $t$ and $v$. Thus Lemmas 3.1 and 3.2 apply uniformly over $v \in \mathfrak{V}_\Lambda$. For costs, on $B_R$,

$$\begin{aligned} &|c_n(x, v(t,x)) - c_0(y, v(t,y))| \\ &\qquad \le \delta_n(R) + \sup_{\substack{|x'|\le R, |y'|\le R \\ |x'-y'|\le r,\ |a-a'|\le \Lambda r}} \left|c_0(x',a) - c_0(y',a')\right|, \end{aligned}$$

with $r = |x-y|$. The last modulus tends to zero as $r \downarrow 0$ because $c_0$ is continuous on the compact set $B_R \times A$. The proofs of Lemma 3.3 and Theorems 3.4–4.1 then go through verbatim. □

The regularity assumption in Corollary 5.1 is not a technical artifact of the proof. Without some condition of this kind, even deterministic degenerate dynamics can make feedback costs discontinuous.

**Proposition 5.2** (Discontinuous feedback obstruction)**.** *State convergence under a fixed feedback does not imply cost convergence for arbitrary measurable feedbacks.*

*Proof.* Let $d = 1$, take $A = \{0, 1\}$ with the discrete topology, set $\sigma_n \equiv 0$, and let

$$b_n(x, a) = -x + \frac{1}{n} \quad (n \geq 1), \qquad b_0(x, a) = -x.$$

Let $c_n = c_0$ be given by $c(x, a) = a$. The coefficients satisfy the global Lipschitz and locally uniform convergence assumptions, and the costs are bounded and continuous on $\mathbb{R} \times A$.

Use the Borel feedback

$$v(x) = \mathbf{1}_{\{x>0\}}, \qquad v(0) = 0.$$

Starting from $x = 0$, the closed-loop state equations are independent of the action and have solutions

$$X_t^n = \frac{1}{n}(1 - e^{-t}) > 0 \quad (t > 0), \qquad X_t^0 = 0.$$

Thus $\sup_{t \leq T} \left| X_t^n - X_t^0 \right| \to 0$ for every $T$, but

$$c(X_t^n, v(X_t^n)) = 1 \quad (t > 0), \qquad c(X_t^0, v(X_t^0)) = 0.$$

Consequently,

$$\int_0^T c(X_t^n, v(X_t^n))\, dt = T, \qquad \int_0^T c(X_t^0, v(X_t^0))\, dt = 0.$$

The finite-horizon cost gap does not vanish. This shows that convergence of states is insufficient unless one also controls how the feedback reacts to small state perturbations, or uses some other compactness/relaxation mechanism. □

# 6 Average cost under explicit uniform stability

Average-cost robustness requires long-run stability in addition to finite-time continuity. The following theorem isolates a sufficient condition. It is deliberately formulated abstractly, because proving the required stability for large degenerate policy classes is itself part of the difficult open problem.

Let $\mathcal{V}$ be a class of stationary or time-homogeneous feedback policies for which the closed-loop processes are well-defined. For $v \in \mathcal{V}$, define

$$A_{T,n}(x, v) := \frac{1}{T} \mathbb{E} \int_0^T c_n(X_t^{n,x,v}, v(X_t^{n,x,v}))\, dt.$$

Assume that for each $n$ and $v$ there is a number $\rho_n(v)$, independent of $x$, intended to be the long-run average cost.

**Assumption 6.1** (Average-cost lifting hypotheses)**.** *The following hold.*

*(E1) For every compact $K \subset \mathbb{R}^d$ and $T < \infty$,*

$$\sup_{x \in K} \sup_{v \in \mathcal{V}} |A_{T,n}(x, v) - A_{T,0}(x, v)| \longrightarrow 0.$$

*(E2) For every compact $K \subset \mathbb{R}^d$,*

$$r_K(T) := \sup_{n \geq 0} \sup_{v \in \mathcal{V}} \sup_{x \in K} |A_{T,n}(x, v) - \rho_n(v)| \longrightarrow 0 \qquad \textit{as } T \to \infty.$$

Define $\rho_n^* := \inf_{v \in \mathcal{V}} \rho_n(v)$.

**Theorem 6.2** (Average-cost robustness under uniform stability). *Under Assumption 6.1,*

$$\sup_{v\in\mathcal{V}} |\rho_n(v) - \rho_0(v)| \longrightarrow 0, \qquad \rho_n^* \longrightarrow \rho_0^*.$$

*If $v_n \in \mathcal{V}$ satisfies $\rho_n(v_n) \le \rho_n^* + \varepsilon_n$ with $\varepsilon_n \downarrow 0$, then*

$$\rho_0(v_n) - \rho_0^* \longrightarrow 0.$$

*Proof.* Fix an arbitrary initial point $x$ and use $K = \{x\}$. For every $T < \infty$,

$$\sup_{v\in\mathcal{V}} |\rho_n(v) - \rho_0(v)| \le 2r_{\{x\}}(T) + \sup_{v\in\mathcal{V}} |A_{T,n}(x,v) - A_{T,0}(x,v)|\,.$$

Take $\limsup_{n\to\infty}$ and use (E1); then let $T \to \infty$ and use (E2). The convergence of $\rho_n^*$ follows from the infimum inequality. Finally,

$$\begin{aligned} 0 \le \rho_0(v_n) - \rho_0^* &\le |\rho_0(v_n) - \rho_n(v_n)| + \big(\rho_n(v_n) - \rho_n^*\big) + |\rho_n^* - \rho_0^*| \\ &\le \sup_{v\in\mathcal{V}} |\rho_0(v) - \rho_n(v)| + \varepsilon_n + |\rho_n^* - \rho_0^*| \longrightarrow 0. \end{aligned}$$

□

**Corollary 6.3** (Regular-feedback average-cost partial progress). *Let $\mathcal{V}$ be a stationary version of the regular feedback class from Section 5. If the uniform ergodic approximation condition (E2) holds for this class, then Theorem 6.2 applies. Hence average costs and near-optimal regular stationary feedbacks are robust within $\mathcal{V}$.*

*Proof.* Condition (E1) follows from the finite-time cost convergence proved in Corollary 5.1, divided by $T$. Condition (E2) is assumed. Apply Theorem 6.2. □

# 7 Nondegenerate approximations of a degenerate limit

The theorem includes the common case in which the approximate models are nondegenerate but the true model is not.

**Corollary 7.1** (Vanishing-noise approximations). *Let $(b_0, \sigma_0, c_0, g_0)$ satisfy the assumptions above, possibly with singular or identically zero $\sigma_0\sigma_0^\top$. On a space carrying an $(m+d)$-dimensional Brownian motion, define*

$$\Sigma_0(x,a) := \big(\sigma_0(x,a) \;\; 0_{d\times d}\big), \qquad \Sigma_n(x,a) := \big(\sigma_0(x,a) \;\; n^{-1/2} I_d\big),$$

*with $b_n = b_0$, $c_n = c_0$, and $g_n = g_0$. Then*

$$\Sigma_n(x,a)\Sigma_n(x,a)^\top = \sigma_0(x,a)\sigma_0(x,a)^\top + n^{-1} I_d,$$

*so each approximate model is nondegenerate, while the limit may be degenerate. The finite-horizon and discounted robustness conclusions above hold.*

*Proof.* The coefficients $\Sigma_n$ satisfy the same Lipschitz and growth estimates as $\Sigma_0$ up to a harmless uniform additive constant, and

$$\sup_{x,a} \|\Sigma_n(x,a) - \Sigma_0(x,a)\| = n^{-1/2}\, \|I_d\| \to 0.$$

Apply Theorems 3.4 and 4.1. □

## 8 Conclusion

The results establish a degenerate-diffusion robustness theorem for finite-horizon and discounted open-loop controls, and for uniformly regular feedback classes. Thus the part of the robustness mechanism based on finite-time cost continuity survives complete degeneracy of the limiting diffusion. This is genuine partial progress toward the degenerate direction suggested by Pradhan and Yüksel.

It is not a complete solution of their problem. The proof avoids the hardest issue: optimal policies in the original theory may be merely measurable Markov or stationary Markov selectors. In a degenerate system, the dynamics need not smooth out discontinuities of such selectors, and Proposition 5.2 shows that the problem is real even for deterministic ODE limits. A full solution would need a substitute for the elliptic-HJB regularity/selection machinery used in the nondegenerate theory, or a relaxed-control/topological argument strong enough to control the implementation of approximate-model selectors on the true degenerate dynamics.

## References

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